Saikat Das

Full-Stack Developer | AI/ML Engineer

Dhaka, Bangladesh   saikatdasmain47@gmail.com   +880 1790 070535
Portfolio   GitHub   LinkedIn   Codeforces

Professional Summary

Computer Science graduate and backend developer experienced in building and deploying Node.js, Django, REST APIs, and data pipelines on AWS, Vercel, Render, and Firebase. Strong grounding in DSA and system design, also active in AI/ML research and engineering.

Technical Skills

Backend	Node.js, Express.js, Django, MVC
Databases & Cloud	MongoDB, MySQL, Firebase, PostgreSQL, AWS, Cloudinary, Vercel, Render
Frontend	React.js (Vite), Tailwind CSS, EJS, Bootstrap
Languages	C++, Python, JavaScript, MATLAB, Java (basic), PHP (basic), SQL
Tools & Concepts	Git/GitHub, Docker (basic), Linux, Jupyter, OS, Networks, System Design, OOP, DSA
AI / ML / DL	PyTorch, TensorFlow/Keras, Scikit-learn, Transformers, Pandas, NumPy, NLP

Education

B.Sc. in Computer Science and Engineering — Graduated 2026

Ahsanullah University of Science and Technology (AUST), Dhaka
CGPA: 3.32 / 4.00 | 3.65 in last 80 credits

HSC - Chittagong College (2020) — GPA 5.00/5.00
SSC - Saint Placid's High School (2018) — GPA 5.00/5.00

Full-Stack Projects

CommerceX — Multi-Vendor Marketplace API

GitHub | Live Demo

Stack: Django, DRF, PostgreSQL, Redis, Groq, JWT

- Architected an enterprise e-commerce backend with role-based access (Customer/Seller/Admin), implementing atomic `select_for_update()` locking to guarantee 100% inventory accuracy and prevent overselling under concurrent load.
- Engineered a hybrid AI shopping assistant combining deterministic PostgreSQL full-text search with grounded LLM responses (Groq), eliminating product hallucination while leveraging Redis caching to slash latency and reduce API costs.
- Orchestrated complex business workflows including per-line fulfillment state machine, seller approval queues, immutable audit logging (soft-delete), and price-snapshot cart system that proactively flags real-time changes.

NovaShop — AI-Powered E-Commerce Platform

GitHub

Stack: React, Node.js, Express, MongoDB, TF-IDF, JWT

- Engineered lightweight in-memory TF-IDF vector index with cosine similarity for zero-latency recommendations; aggregated purchase/wishlist/browsing history for personalized suggestions with cold-start fallback.
- Architected hybrid semantic search with substring boosts and virtual assistant parsing local intent before optionally calling LLMs (Groq/Ollama)—implementing graceful degradation for 100% uptime.

- Developed full-stack MERN platform with JWT role-based access, secure inventory validation, verified-purchase-only review gates, and responsive dark-mode UI with context-based global state.

PingMe — Real-Time Chat Application

GitHub

Stack: React, Node.js, Express, MongoDB, Socket.io, Cloudinary

- Engineered full-duplex real-time messaging using Socket.io with persistent WebSocket connections for instant 1-on-1 messages and live online/offline presence indicators—ensuring sub-second delivery without polling overhead.
- Integrated Cloudinary for asynchronous image uploads, decoupling media processing from the real-time pipeline to prevent latency spikes.
- Built responsive UI using Tailwind CSS and Vite, optimizing client-side state management for chat histories, unread counts, and connection status.

Paira-A Messenger — Native Android Chat App

GitHub

Stack: Java, Android SDK, Firebase Realtime DB, Firebase Auth

- Developed native Android messaging app leveraging Firebase Realtime Database listeners for persistent bi-directional data streams, enabling instant delivery and live presence without custom WebSocket servers.
- Integrated Firebase Authentication and asynchronous media-sharing pipelines via Firebase Storage, managing memory constraints and file-permission policies.
- Architected UI/UX using XML layouts and Java lifecycle-aware components, optimizing for background/foreground state transitions.

Hotel Management System — Dynamic Booking Platform

GitHub

Stack: PHP, MySQL, Bootstrap, JavaScript, XAMPP

- Engineered secure, server-side rendered hotel booking engine using raw PHP and MySQL with session-based auth, RBAC, and atomic transactions to prevent double-booking.
- Designed normalized relational schema with foreign key constraints for users, rooms, bookings, payments, and inquiries—maintaining referential integrity.
- Built responsive, mobile-first front-end using Bootstrap with dynamic admin dashboards for real-time booking oversight and operational reporting.

AI/ML Engineering & Research Projects

SentimentSphere — ML Sentiment Classifier

GitHub

Stack: Python, Scikit-learn, BeautifulSoup, Pandas, NumPy

- Built binary sentiment classifier using Logistic Regression on a 3.6M-label e-commerce review dataset with preprocessing pipeline (stopword removal, stemming, noise filtering).
- Engineered live data ingestion pipeline using BeautifulSoup/Requests to scrape real-time reviews with dynamic error detection and model update mechanisms.
- Deployed CLI/UI interface for instant testing, serializing model and vectorizer ('.pkl') for production-ready inference.

Edge-AI Waste Classification — Real-Time IoT Smart Bin

GitHub | Demo Video

Stack: Python, TensorFlow Lite, Raspberry Pi, OpenCV, Firebase, Flask

- Architected edge-AI system on Raspberry Pi 4B, benchmarking 6 CNN architectures using custom Edge Deployment Score (EDS) balancing accuracy, throughput, model size, latency, CPU/RAM, and thermal

performance.
- Engineered complete IoT pipeline integrating FPM10A fingerprint biometrics, stepper/servo motor actuation, HC-SR04 fill-level monitoring, and IP camera capture—all with sub-500ms inference latency.
- Implemented tamper-evident cryptographic ledger with hash-chaining and digital signatures for secure logging, paired with Firebase cloud sync for remote monitoring.

Publications Under Review

Saikat Das, et al. "A Deployment and Resource Aware Edge AI Framework for Real-Time Waste Classification with Biometric Authentication and Secure Logging" - Under review, submitted April 2026

Research Interests: Computer Vision, Health AI, Efficient Deep Learning, Edge AI / TinyML, Vision Transformers, Vision-Language Models, NLP, Model Quantization & Compression, Hardware-Aware NAS.

Certifications

- **MathWorks:** MATLAB Onramp, ML Onramp, DL Onramp — Jul 2026
- **OpenCV University:** VLM Bootcamp, TF-Keras, PyTorch, Python — 2025–26
- **Google Cloud:** Prompt Design, Responsible AI, Intro to LLMs, Intro to GenAI, ML Engineer Guide 2025–26
- **IBM:** Introduction to Data Engineering — Feb 2026
- **HP LIFE / Coursera:** Professional Networking, WordPress Website — 2026

Achievements & Additional Information

- Solved **200+ problems** on competitive programming platforms (Codeforces, LeetCode).
- Reviewer activity for undergraduate research symposium (AUST CSE Fest 2024); graduate-level coursework in Advanced Machine Learning and Digital Image Processing.
- **Languages:** English - B1 (Intermediate); Bengali - Native (C2).
- **Interests:** BackendDev, AI for sustainability, efficient deep learning, multimodal reasoning, open-source scientific contribution.

References

Ms. Nusrat Jahan
Assistant Professor, Dept. of CSE, AUST
nusratjahan.cse@aust.edu

Mr. Mustofa Ahmed
Lecturer, Dept. of CSE, AUST
mustofa.cse@aust.edu